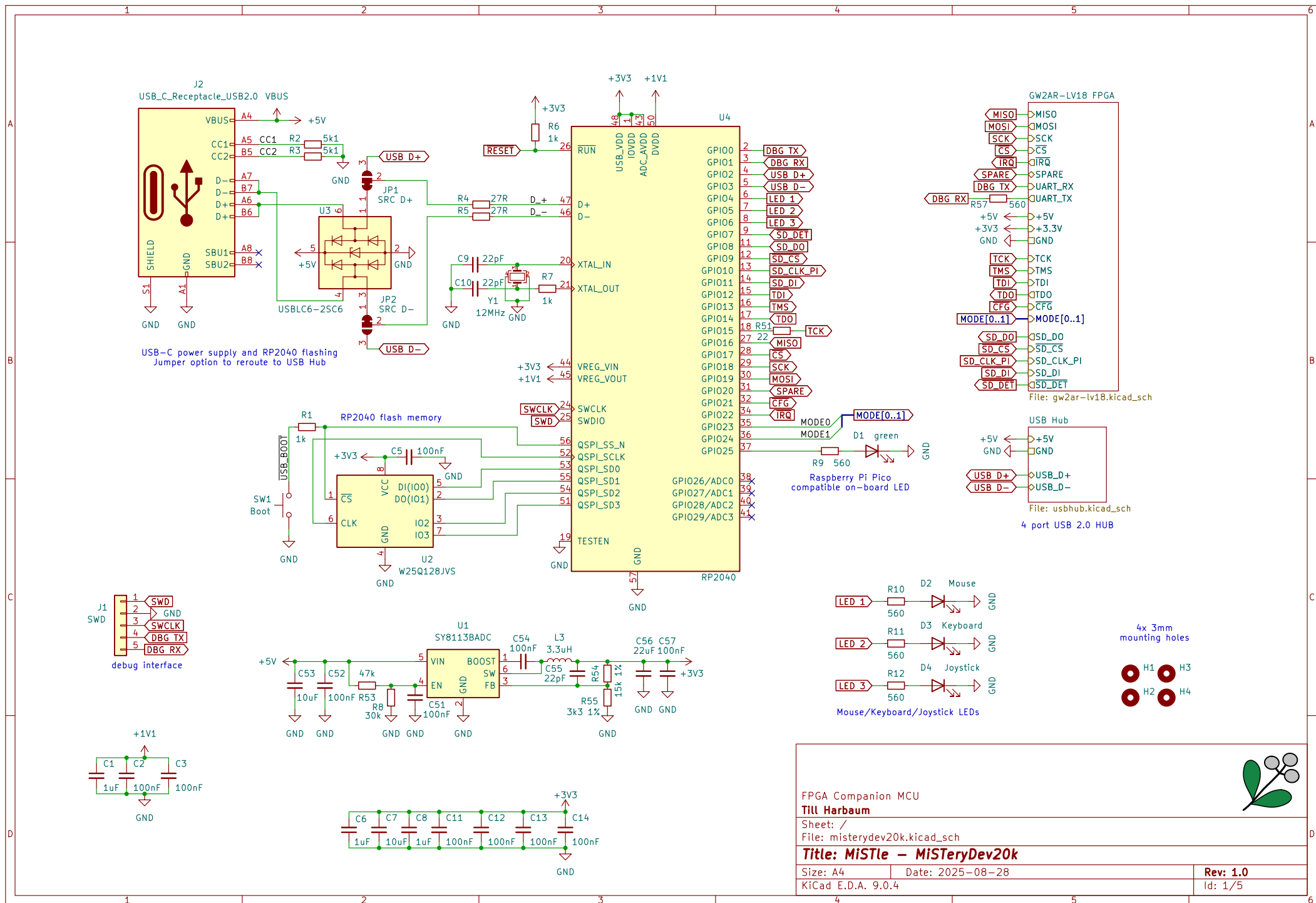
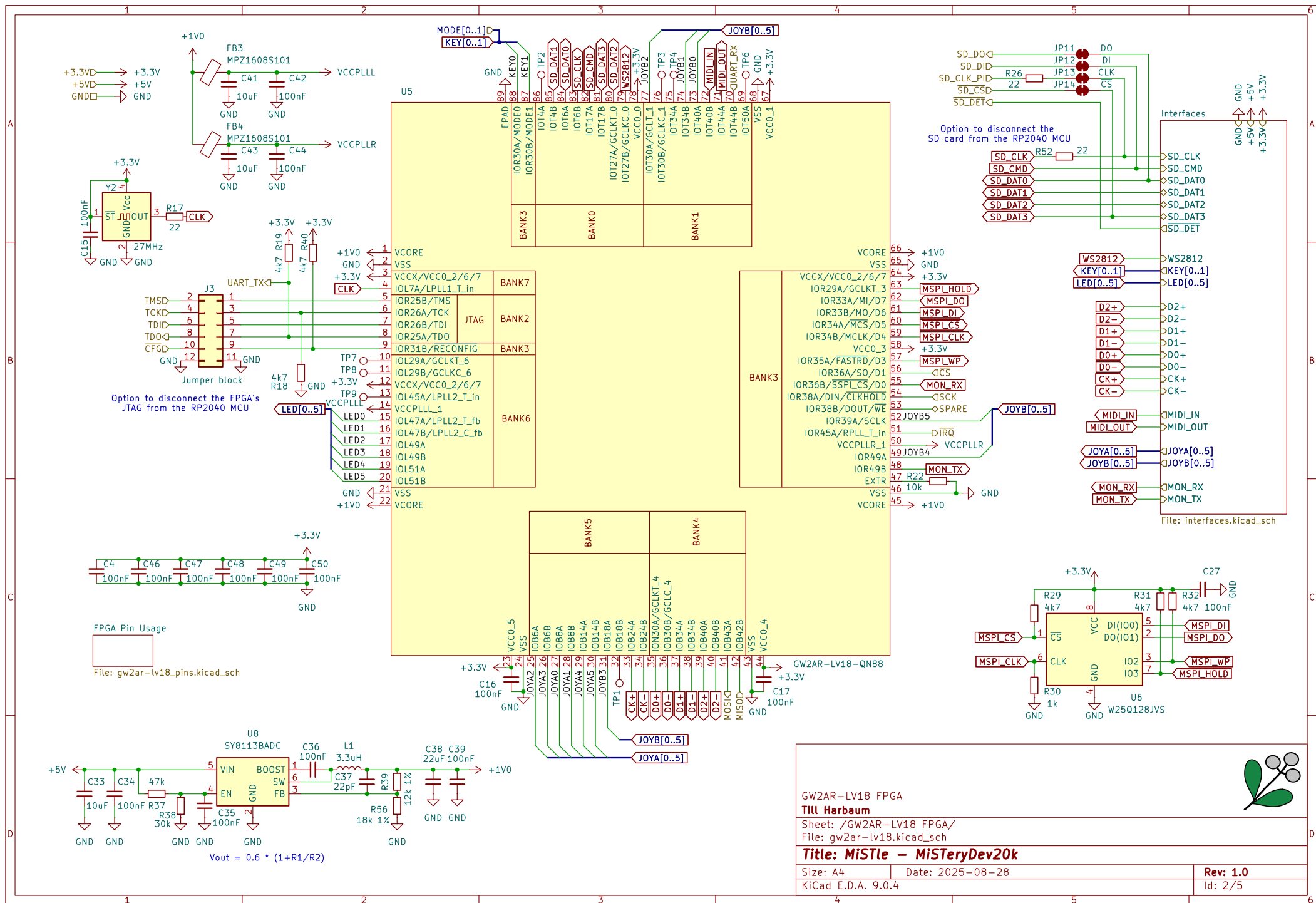
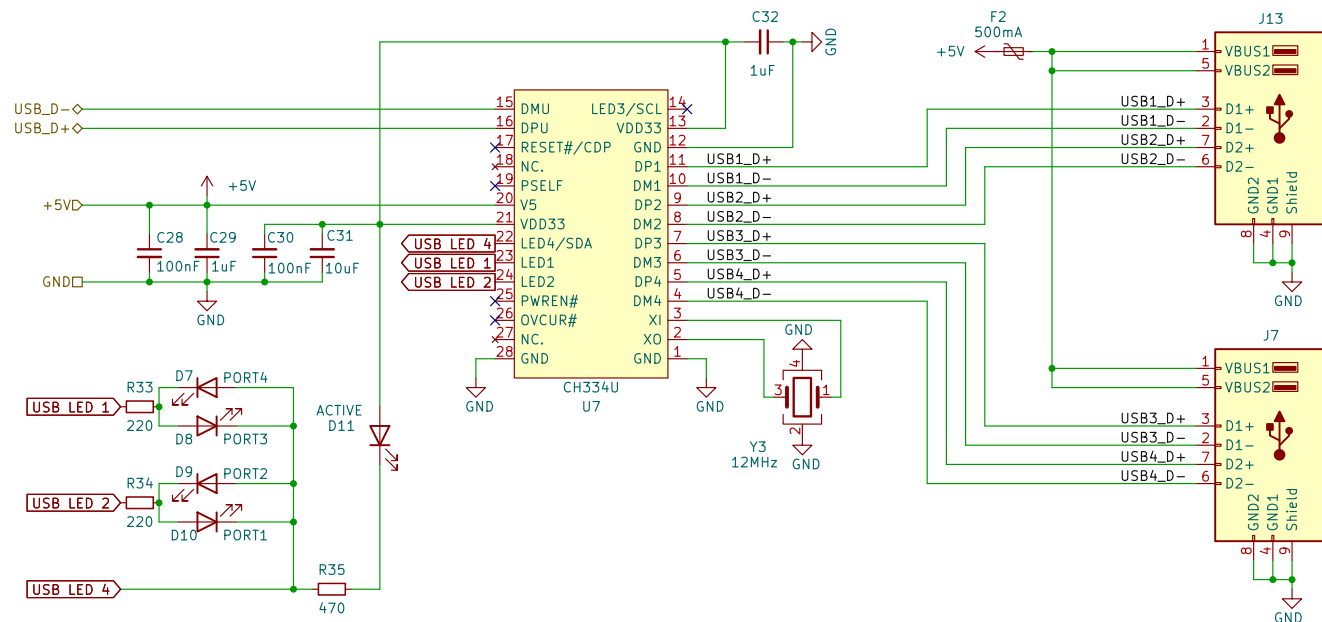








USB Hub

Stefan Voss, Till Harbaum

Sheet: /USB Hub/

File: usbbhub.kicad_sch

Title: MiSTle – MiSTeryDev20k

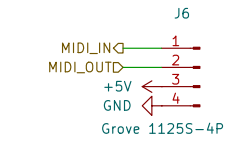
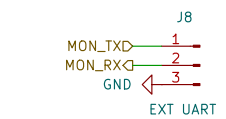
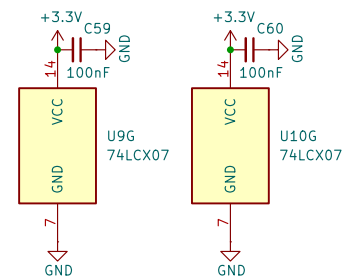
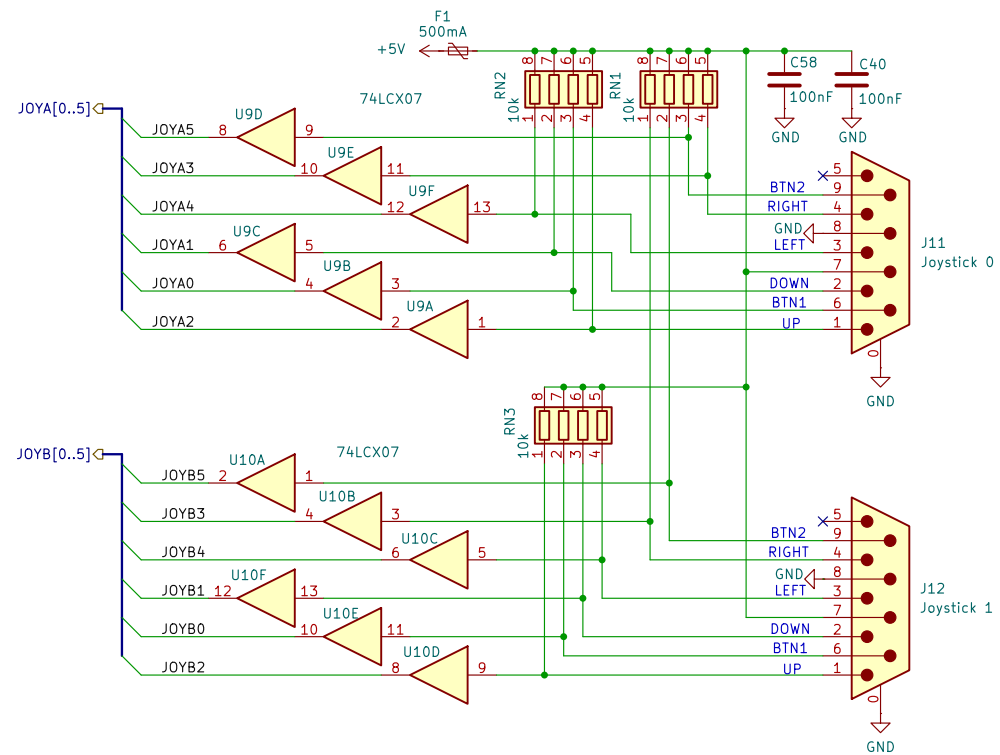
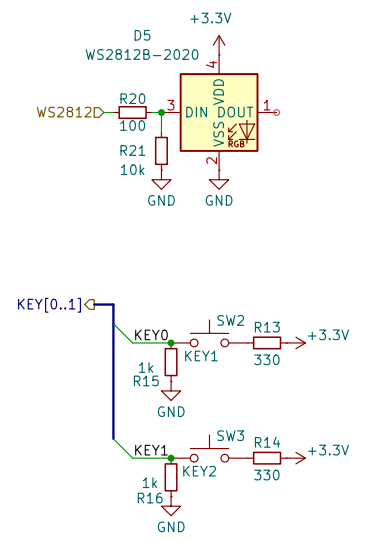
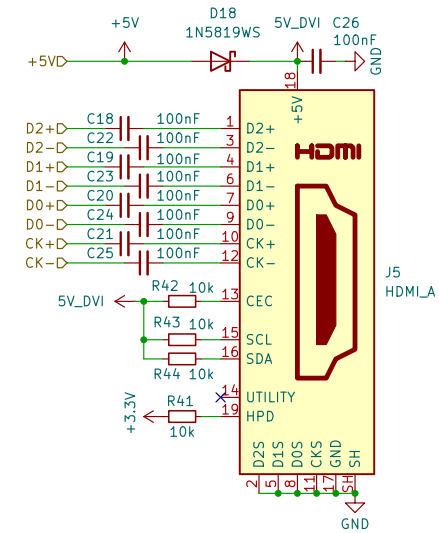
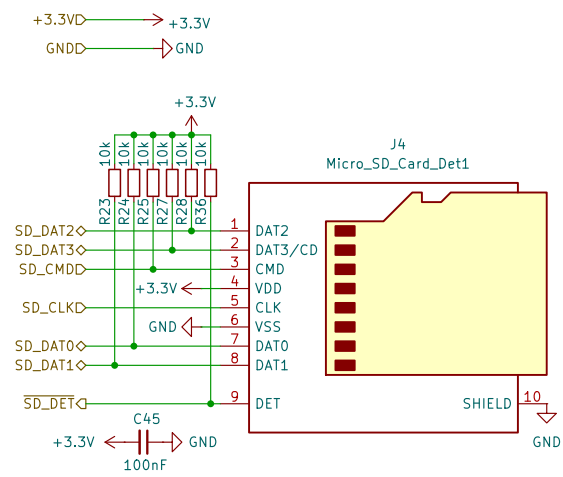
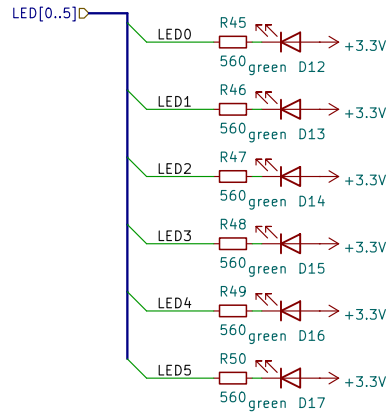
Size: A4 Date: 2025-08-28

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to be used with e.g.
M5 SAM2695 MIDI, or
M5 SAM2695 SYNTH

Peripherals	
Till Harbaum	
Sheet: /GW2AR-LV18 FPGA/Interfaces/	
File: interfaces.kicad_sch	
Title: MiSTle – MiSTeryDev20k	
Size: A4	Date: 2025-08-28
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	Id: 4/5



Pin	Function	TN20k (yellow=on hdr)	NanoMig	Usage	Pin	Function	TN20k (yellow=on hdr)	NanoMig	Usage
1	Vcore				45	Vcore			
2	Vss				46	Vss			
3	Vccc				47	EXTR			
4	IOL7A	SYS_CLK	CLOCK		48	IOR49B	LCD_DE	MON_TX	EXT UART OUT
5	IOR25B	SYS_TMS			49	IOR49A	LCD_BL	SPARE4	JS1-3 LEFT
6	IOR26A	SYS_TCK			50	Vccpll			
7	IOR26B	SYS_TDI			51	IOR45A	PA_EN	MOS4	
8	IOR25A	SYS_TDO	UART_TX	FPGA->MCU	52	IOR39A	EDID_DAT	SPARE5	JS1-9 BTN2
9	IOR31B	RECFG			53	IOR38B	EDID_CLK	MOS5	
10	IOL29A	5351_CKP			54	IOR38A	I2S_DIN	MOS3	
11	IOL29B	5351_CKN			55	IOR36B	I2S_LRCK	MON_RX	EXT UART IN
12	Vccx				56	IOR36A	I2S_BCLK	MOS2	
13	IOL45A	HSPI_SCLK	BL616_SCLK		57	IOR35A	MSPI_WP	MSPI_WP	
14	Vccpll				58	Vcc			
15	IOL47A	LED0	LED0		59	IOR34B	MSPI_CLK	MSPI_CLK	
16	IOL27B	LED1	LED1		60	IOR34A	MSPI_CS	MSPI_CS	
17	IOL49A	LED2	LED2		61	IOR33B	MSPI_DI	MSPI_DI	
18	IOL49B	LED3	LED3		62	IOR33A	MSPI_DO	MSPI_DO	
19	IOL51A	LED4	LED4		63	IOR29A	MSPI_HOLD	MSPI_HOLD	
20	IOL51B	LED5	LED5		64	Vccx			
21	Vss				65	Vss			
22	Vcore				66	Vcore			
23	Vcc				67	Vcc			
24	Vss				68	Vss			
25	IOB6A	LCD_HS	IO2	JS0-1 UP	69	IOT50A	SYS_TX		
26	IOB6B	LCD_VS	IO3	JS0-4 RIGHT	70	IOT44B	SYS_RX	UART_RX	MCU->FPGA
27	IOB8A	LCD_B7	IO0	JS0-6 BTN1	71	IOT44A	HSPI_DIN0	MIDI_OUT	
28	IOB8B	LCD_B6	IO1	JS0-2 DOWN	72	IOT40B	HSPI_DIN1	MIDI_IN	
29	IOB14A	LCD_B5	IO4	JS0-3 LEFT	73	IOT40A	HSPI_DIN2	SPARE0	JS1-6 BTN1
30	IOB14B	LCD_B4	IO5	JS0-9 BTN2	74	IOT34B	HSPI_DIN3	SPARE1	JS1-2 DOWN
31	IOB18A	LCD_B3	SPARE3	JS1-4 RIGHT	75	IOT34A	HSPI_DIR	BL616_DIR	
32	IOB18B	LCD_G7			76	IOT30B	HSPI_DAT	BL616_DAT	
33	IOB24A	DVI_CKP	DVI_CKP		77	IOT30A	LCD_CLK	SPARE2	JS1-1 UP
34	IOB24B	DVI_CKN	DVI_CKN		78	Vcc			
35	IOB30A	DVI_D0P	DVI_D0P		79	IOT27B	WS2812	WS2812	
36	IOB30B	DVI_D0N	DVI_D0N		80	IOT27A	SDIO_D2	SDIO_D2	
37	IOB34A	DVI_D1P	DVI_D1P		81	IOT17B	SDIO_D3	SDIO_D3	
38	IOB34B	DVI_D1N	DVI_D1N		82	IOT17A	SDIO_CMD	SDIO_CMD	
39	IOB40A	DVI_D2P	DVI_D2P		83	IOT6B	SDIO_CLK	SDIO_CLK	
40	IOB40B	DVI_D2N	DVI_D2N		84	IOT6A	SDIO_D0	SDIO_D0	
41	IOB43A	LCD_R4	MOS0		85	IOT4B	SDIO_D1	SDIO_D1	
42	IOB42B	LCD_R3	MOS1		86	IOT4A	HSPI_CS	BL616_CS	
43	Vss				87	IOR20B	KEY2	BTN_RESET	
44	Vcc				88	IOR30A	KEY1	BTN_USER	

FPGA pin usage

Till Harbaum

Sheet: /GW2AR-LV18 FPGA/FPGA Pin Usage/

File: gw2ar-lv18_pins.kicad_sch

Title: MiSTle - MiSTeryDev20k

Size: A4 | Date: 2025-08-28

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